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The Metaverse as a Game Ecosystem: Evaluating Player Interaction Through Mini-Game Integration and Experience Analysis

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Abstract*—The metaverse represents a rapidly evolving digital frontier, offering unprecedented opportunities to blend immersive environments with interactive gaming elements. This study explores the integration of three mini-games into a single-player metaverse designed as a 3D adventure island, showcasing the metaverse as a dynamic game ecosystem. These mini games are embedded seamlessly within the virtual environment, creating an engaging and cohesive user experience. To evaluate the impact of this integration on player experience, the study employs the Game Experience Questionnaire (GEQ), a robust framework for assessing dimensions such as immersion, engagement, and enjoyment. GEQ results provide valuable insights into player experiences, with the Positive Experience dimension achieving the highest score of 1.93, reflecting strong satisfaction and a sense of accomplishment during gameplay. The tiredness dimension scored 0.79, indicating minimal physical or mental fatigue, while negative affect scored just 0.63, suggesting very low levels of frustration or regret. A moderate score of 1.35 in the return to reality dimension highlights the immersive nature of the experience, with some players requiring time to reorient post-game. These findings demonstrate the effectiveness of mini games in enhancing engagement and enjoyment while maintaining a balanced gameplay experience.

***Keywords*—metaverse, game, game ecosystem, player experience, game experience questionnaire, immersive**

I. Introduction

The metaverse, a continuous and immersive virtual environment, has arisen as a revolutionary notion in digital entertainment and interactive experiences. The metaverse integrates technologies like augmented reality (AR), virtual reality (VR), and sophisticated game design, providing unparalleled prospects for creativity, social interaction, and user engagement [1]. The metaverse, as a digital ecosystem, is not just a technology framework but a dynamic environment that accommodates many activities such as gaming, education, and social networking [2]. As a synthesis of many technologies, it offers a platform for the integration of multiple game aspects that both entertain and cultivate distinctive experiences for users[3].

In recent years, the gaming industry has increasingly adopted the metaverse as a platform for innovation [4]. Unlike traditional video games, where players are confined to predefined narratives and environments [5], metaverse-based games emphasize open-world exploration, real-time interaction, and player-driven content [6]. These characteristics align with the growing demand for personalized and shared gaming experiences, marking a paradigm shift in how games are designed, played, and monetized [4]. Furthermore, the integration of games within metaverse ecosystems has gained traction as a strategy to enhance player engagement, diversify gameplay, and cater to a wide range of player preferences [7].

Games serve as modular, self-contained activities that complement the larger metaverse experience [3]. They offer opportunities for players to test their skills, collaborate, and compete in scenarios distinct from the primary game world[8]. This modular approach not only enriches the gaming experience but also provides valuable insights into player behaviour and preferences [9]. By analyzing player interactions within games, researchers and developers can better understand factors influencing engagement, satisfaction, and overall game experience [10].

This article explores the integration of three mini-games into a single-player metaverse conceptualized as a 3D adventure island. The mini games are intricately integrated into the environment, highlighting immersion, fluidity, and interactivity. The research examines how this integration improves player experience, satisfaction, and engagement.

II. Literature Review

A. Gaming in the Metaverse

Gaming in the metaverse represents a significant evolution in digital interaction, blending immersive virtual environments

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with social and economic elements [3]. The metaverse allows for a seamless integration of gaming experiences across various platforms, enhancing user engagement and interaction [11]. Research indicates that the social aspect of metaverse gaming is crucial, as it fosters community building and collaborative play, which are essential for player retention and satisfaction [12]. The integration of blockchain technology in metaverse gaming has introduced new paradigms such as play-to-earn models, where players can earn real-world value through in-game activities [13]. This not only enhances player motivation but also creates a sustainable economic ecosystem within the virtual world. Additionally, the use of non-fungible tokens (NFTs) allows for true ownership of digital assets, further enriching the gaming experience [13].

Studies have also highlighted the importance of user-generated content in the metaverse, which empowers players to create and share their own game elements, thereby increasing engagement and personalization [12]. The flexibility and interoperability of game assets across different virtual environments contribute to a more cohesive and immersive experience [11]. Overall, the metaverse is poised to revolutionize the gaming industry by offering unparalleled levels of immersion, social interaction, and economic opportunities for players.

B. Evaluating Player Expericene

Evaluating player experience in the metaverse involves understanding how users interact with and perceive virtual environments [14]. Research has shown that user experience (UX) in the metaverse is influenced by factors such as immersion, usability, and social interaction. Immersion is critical, as it enhances the sense of presence and engagement, making players feel as though they are truly part of the virtual world [15]. Usability, including intuitive controls and seamless navigation, significantly impacts player satisfaction and retention [11].

Social interaction is another key component, as it fosters a sense of community and collaboration among players. Studies on platforms like ROBLOX have highlighted the importance of user control and content engagement in enhancing the overall experience [12]. Additionally, the customization of avatars and environments plays a vital role in personalizing the experience and increasing player attachment to the virtual world[16].

III. Research Methodology

The research is conducted within a meticulously designed metaverse island, crafted as a single-player virtual environment to ensure controlled and focused gameplay experiences. This island features an outdoor setting that encourages exploration, thereby promoting player navigation and interaction. At the center of the island is an interactive building, which serves as the gateway to a series of integrated mini games. This design facilitates a seamless transition from unrestricted exploration to structured gameplay. Figure 1 illustrates the game flow in the metaverse mini-game integration.

This study aims to evaluate player interactions within the metaverse environment and their engagement with mini games. Participants are required to navigate the island, locate the central structure, and complete all mini games to progress.

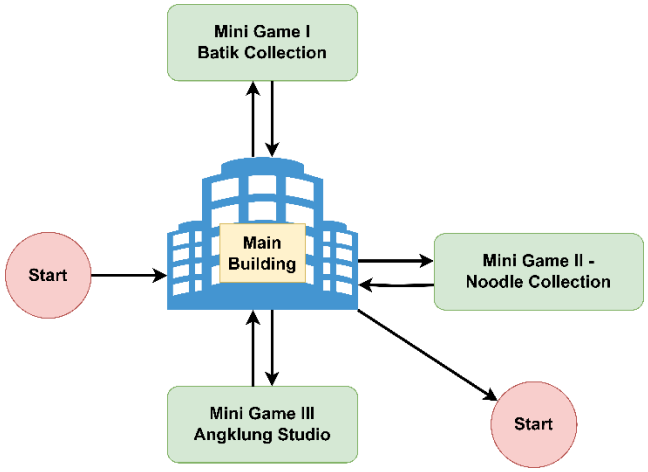

Fig 1. Game Flow in Metaverse Mini Game Integration

Upon completion of the gameplay, player experiences are rigorously assessed using the Game Experience Questionnaire (GEQ). This evaluation focuses on key aspects of the player experience, such as engagement, challenge, and satisfaction, providing valuable insights into the effectiveness of the metaverse as a gaming environment [17].

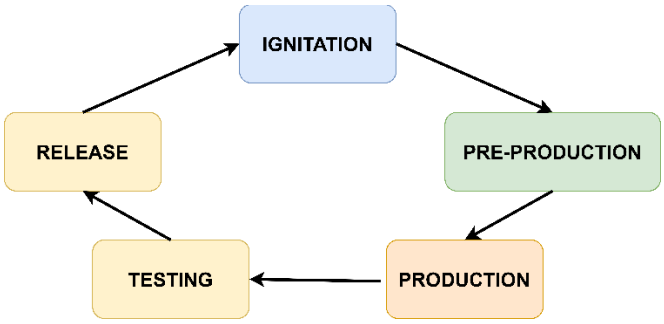

Fig 2. Game Development Life Cycle for Integration mini game through Metaverse

The development of the metaverse environment follows the Game Development Life Cycle (GDLC) framework shown in figure 2 prioritizing player-centric methodologies. The design phase begins with initialization, pre-production, production and testing which includes the following phases shown in Table 1.

TABLE I. Game Development Life Cycle Phases for Integration Mini Game Through Metaverse

No.	Phases	Approaching	Output
1.	Initiation	Rough Concept	Game Concept Documents
2.	Pre-production	Focus Group Discussion	Game Design Documents
3.	Production	Prototype	Game Prototype
4.	Testing	Alpha Testing	Game Prototype & Feedback
		Beta Testing	Game Prototype & Feedback
5.	Release	Build on platform	Release on web platform
		GEQ	Result of player experience

The game development life cycle for integrating mini games into a metaverse ecosystem consists of five structured phases:

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initiation, pre-production, production, testing, and release. The process begins with the initiation phase, where a rough concept is formulated, leading to the creation of game concept documents. These documents emphasize incorporating Indonesian local content as the central theme for the mini games, aiming to showcase and promote Indonesia's rich cultural heritage to an international audience. The pre-production phase follows, during which focus group discussions are conducted to refine initial ideas and develop detailed Game Design Documents (GDD). The GDD outlines the design of three culturally inspired mini games. The first mini-game, *Batik Collection*, requires players to collect several unique batik patterns, with informational content about each pattern displayed upon collection. The second mini-game, *Noodle Collection*, challenges players to gather *Indomie*, Indonesia's iconic instant noodles, within a set time limit. The third mini-game, *Angklung Studio*, allows players to play the *angklung*, a traditional Indonesian musical instrument, using a guided score as a reference.

During the production phase, the mini-game prototype is developed, serving as the foundation for subsequent iterations. We develop using unity game engine and blender as a 3D modeler tool. The testing phase includes alpha and beta testing to evaluate the prototype and gather feedback, ensuring quality improvements. During this phase, the internal development team plays a critical role by conducting thorough testing, identifying potential issues, and refining the gameplay experience. Their input is instrumental in enhancing the overall quality and ensuring the mini games meet the intended design and performance standards.

Finally, in the release phase, the mini game is deployed on the web platform, and player interaction is assessed through the Game Experience Questionnaire (GEQ) to analyze and improve the overall player experience [18] with particular emphasis on the Core Module and Post-Game Module.

TABLE II. THE GEQ CORE MODULE SCORING GUDILINES

No	Component	Statements
1.	Competence	S2, S10, S15, S17, S21
2.	Immersion	S3, S12, S18, S19, S27, S30
3.	Flow	S5, S13, S25, S28, S31
4.	Challenge	S22, S24, S29
5.	Positive Affect	S11, S23, S26, S32, S33
6.	Negative Affect	S7, S8, S9, S16
7.	Tension	S1, S4, S6, S14, S20

The Core Module evaluates immediate, in-game experiences such as immersion, challenge, and enjoyment. As detailed in Table II, this module consists of seven components: Competence (measuring perceived skill and success), Immersion (depth of involvement in the game world), Flow (smoothness and absorption in gameplay), Challenge (perceived difficulty), Positive Affect (feelings of joy and satisfaction), Negative Affect (frustration or disappointment), and Tension (feelings of pressure or nervousness). Each component is calculated based on specific statements (e.g., S2, S10, etc.), and participants rate each item on a scale from 1 to 5. The average score per component is computed by aggregating the responses and dividing by the number of respondents[16].

Meanwhile, the Post-Game Module, as shown in Table III, captures players' reflections after gameplay and includes four components: Positive Experience (overall satisfaction and pride), Tiredness (physical and mental fatigue), Negative Experience (post-game dissatisfaction or regret), and Return to Reality (difficulty in transitioning back to the real world after immersion). This combination of immediate and reflective metrics allows for a comprehensive evaluation of how players perceive and emotionally respond to the gaming experience.

TABLE III. THE GEQ POST-GAME MODULE SCORING GUDILINES

No	Component	Statements
1.	Positive Experience	S1, S5, S7, S8, S12, S16
2.	Tiredness	S2, S4, S6, S11, S14, S15
3.	Negative Experience	S10, S13
4.	Return to Reality	S3, S9, S17

The Social Presence Module was not employed in this study, as it is designed specifically for multiplayer settings. However, future iterations of the metaverse, which aim to include multiplayer features, will incorporate this module to assess how social interactions influence player engagement and experience. This structured evaluation framework highlights the importance of iterative feedback in refining and optimizing mini-game integration within the metaverse ecosystem. This structured life cycle emphasizes iterative feedback and evaluation to optimize the integration and effectiveness of mini games within the metaverse.

IV. RESULT

We develop a single player metaverse to provide players with an immersive environment in which to interact and explore with average playtime duration of approximately 30 minutes. central building that serves as the gateway to a series of engaging mini games is included in the environment, which is designed as an expansive 3D world. The landscapes are detailed. A seamless blend of exploration and gameplay within the virtual space is achieved as players navigate through the environment, uncovering interactive elements and visual cues that enhance the sense of discovery.

To interact with the mini games that are integrated into the metaverse, players must enter the central building. Players transition from the exploratory phase in the outdoor environment to the structured challenges of the mini games through the building, which serves as a critical component of the gameplay. This design guarantees a clear progression in the player's journey, underscoring the significance of discovery and interaction while simultaneously preserving a seamless transition between exploration and gameplay. Figure 3 shown the main building entrance.

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Fig 3. Main Building entrance in metaverse island

The first mini game, set in a virtual batik shop, immerses players in a culturally rich environment designed to highlight the beauty and intricacy of batik patterns. Players are challenged to gather specific batik motifs strategically placed throughout the shop, as depicted in Figure 4. This task not only challenges their observation skills but also introduces them to the symbolic meanings and artistic elements of traditional batik designs. The game mechanics are designed to be intuitive, tasking players with identifying and collecting patterns within a time limit, fostering engagement and creating a sense of urgency, as illustrated in figure 5. Visual and auditory cues, such as traditional batik-making sounds and vibrant color palettes, enhance the immersive experience, fostering a deeper appreciation for this significant cultural art form.

Fig 4. Instructions for Gameplay in the *Batik Collection's* Mini Game

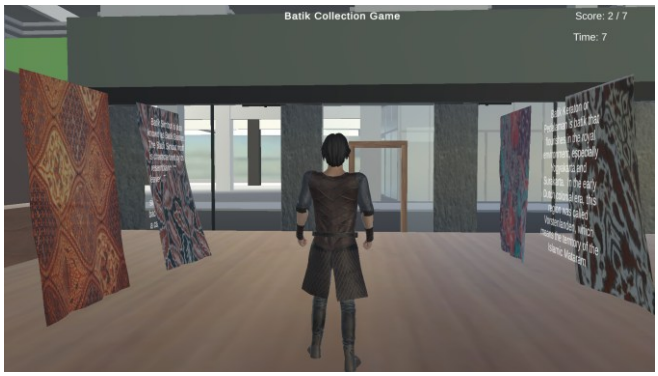

Fig 5. Game mechanics for *Batik Collection's* Mini Game

The second mini-game, *Noodle Collection*, is located to the left of the building entrance and features a dynamic, reflex-based gameplay design. Players must navigate a fast-paced environment to collect noodle packets while avoiding a series of moving obstacles, such as falling debris or rotating barriers. The game mechanics focus on enhancing quick decision-making and hand-eye coordination, requiring players to time their actions accurately for successful gameplay, as illustrated in Figure 6. The setting features the iconic Tugu Station, a renowned train station in Yogyakarta, brought to life with a vibrant market-inspired aesthetic. Bustling sounds and detailed visuals immerse players in the lively atmosphere of this cultural landmark. The combination of action and reflexive challenge ensures that players remain actively engaged, balancing difficulty with enjoyment.

Fig 6. Game Mechanics for *Noodle Collection's* Mini Game

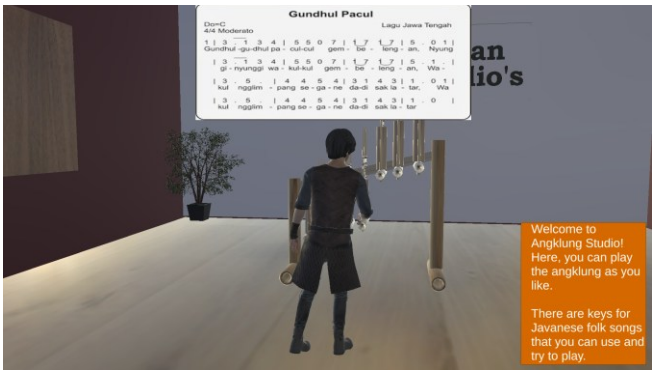

Fig 7. Game mechanics for *Angklung Studio's* Mini Game

The third mini-game, *Angklung Studio*, situated to the right, centers on rhythm and cultural engagement. Players are tasked with playing traditional songs using a virtual Angklung, an Indonesian musical instrument made of bamboo tubes. The game requires players to follow rhythmic patterns displayed on the screen, striking the correct notes in time to create harmonious melodies. The difficulty adjusts dynamically based on player performance, accommodating both beginners and advanced players. The studio environment is designed with a serene aesthetic, featuring traditional decor and calming background music to complement the gameplay. This mini game not only promotes an appreciation of Indonesian musical heritage but also enhances players' sense of rhythm and

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coordination. These detailed designs ensure that each mini game offers a unique and enriching experience, blending gameplay mechanics with cultural elements to provide a well-rounded interaction within the metaverse.

V. Discussion

The game was evaluated on a group of 16 participants, all of whom were male and had an average age of 20 years. The participants were chosen by virtue of their involvement as active students in a game design course, which guaranteed that they were well-versed in the fundamentals of gaming and had a solid understanding of its principles. Furthermore, the participants indicated that they engaged in video game play on a regular basis, rendering them an optimal target audience for assessing the gameplay experience in a metaverse environment. This demographic profile enabled the study to collect insights from individuals who are both knowledgeable and experienced in gaming, thereby providing a well-informed perspective on the game's design, mechanics, and overall player engagement.

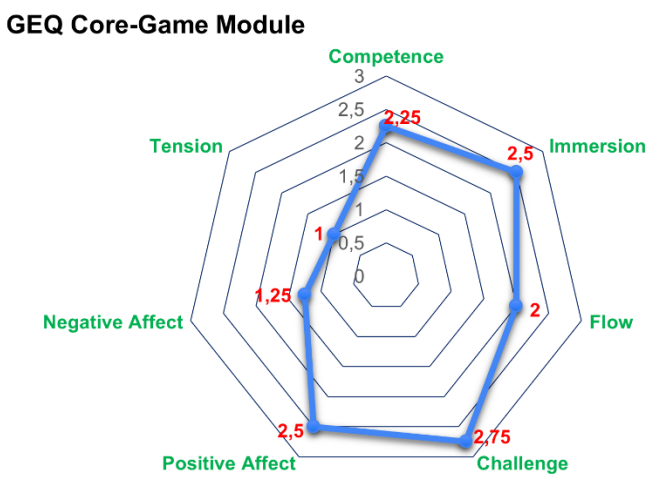

Fig 8. Result of the GEQ Core Game Module

Figure 8 presents the results of the GEQ Core Game Module, highlighting the average scores across various dimensions of the player experience. The highest scores were observed in Challenge (2.75) and Immersion (2.5), indicating that players found the gameplay both engaging and suitably demanding. Similarly, Positive Affect (2.5) received a strong score, reflecting players' enjoyment and satisfaction during gameplay. Competence (2.25) suggests that players felt moderately skilled and capable of managing the tasks presented in the game.

On the other hand, Flow (2.0) reveals opportunities for improvement in delivering a seamless and deeply immersive experience. Meanwhile, Negative Affect (1.25) and Tension (1.0) scored notably low, indicating minimal frustration or negative emotions during gameplay. These findings suggest a well-rounded gaming experience, characterized by high engagement and enjoyment, while also identifying areas for enhancement, particularly in maintaining flow and fostering a greater sense of competence.

Figure 9 presents the results of the GEQ Post-game Module, categorizing scores across various aspects of the player experience following gameplay. Positive Experience achieved the highest average score of 1.93, indicating that players generally felt satisfied, empowered, and proud of their performance. Tiredness scored 0.79, reflecting minimal mental or physical fatigue, suggesting the gameplay was engaging without being overly demanding. Negative Experience recorded a low score of 0.63, signifying very low levels of regret or dissatisfaction, emphasizing the game's effectiveness in delivering a positive emotional experience. Finally, Return to Reality scored 1.35, indicating moderate levels of disorientation post-game but not to the extent of significantly disrupting players' transition back to real-world activities. Overall, the results highlight a favorable reception to the game, characterized by strong positive engagement, minimal fatigue, and a balanced emotional impact.

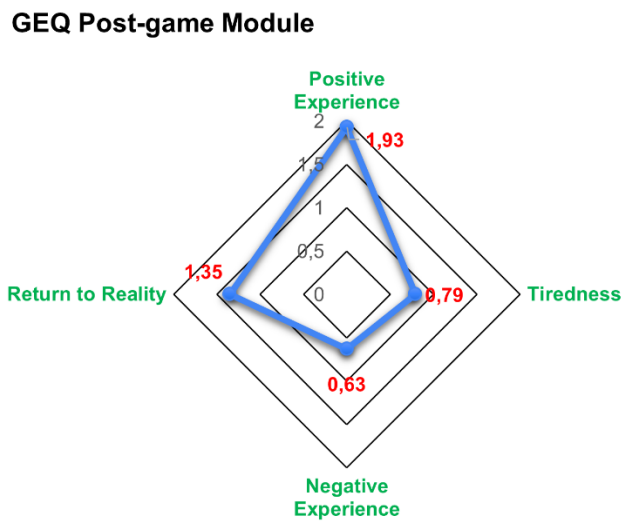

Fig 9. Result of the GEQ Post-game Module

The interrelationship among these dimensions reveals a coherent narrative: the game successfully delivered a challenging and immersive experience that was emotionally positive and minimally taxing. The alignment between Core and Post-Game responses underscores the effectiveness of the game in maintaining consistent user satisfaction across all stages of engagement. This holistic view demonstrates that the metaverse mini-game ecosystem not only entertains but also sustains a well-balanced player experience.

VI. Conclusion And Future Works

This study illustrates the effective integration of three culturally influenced mini games within a single 3D adventure island in the metaverse. The mini games were intricately integrated into the virtual environment, fostering an interesting and unified experience for participants. The outcomes of the Game Experience Questionnaire (GEQ) yield significant insights on player experience. In the Core Game Module, elevated scores in Challenge (2.75) and Immersion (2.5) suggest that participants perceived the gaming as interesting and suitably challenging, whereas Positive Affect (2.5) signifies a sense of enjoyment and fulfilment. Reduced Flow scores (2.0) indicate potential for improving the fluidity of the gaming experience,

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whereas low Negative Affect (1.25) and Tension (1.0) signify negligible frustration or tension during gameplay. The Post-game Module results corroborate these findings, with the Positive Experience component attaining the highest score (1.93), highlighting significant satisfaction and a sense of achievement. Low scores in Tiredness (0.79) and Negative Affect (0.63) reflect minimal physical or mental exhaustion and dissatisfaction, yet a moderate score in Return to Reality (1.35) underscores the immersive quality of the encounter without considerable post-game confusion. These results illustrate the efficacy of the mini games in promoting engagement, enjoyment, and cultural appreciation, while ensuring a balanced and accessible gameplay experience.

Future developments in the metaverse may incorporate multiplayer interactions, facilitating collaborative and competitive gameplay to augment social involvement. Enhanced improvement of game mechanisms, especially to optimize Flow, could yield a more smooth and profoundly immersive experience. The incorporation of customizable difficulty and enhanced cultural elements may engage a wider audience and provide a more tailored gaming experience. Ultimately, longitudinal studies examining the effects of metaverse-based games on cultural comprehension and player retention may yield profound insights into their educational and entertainment significance.

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